Business overview

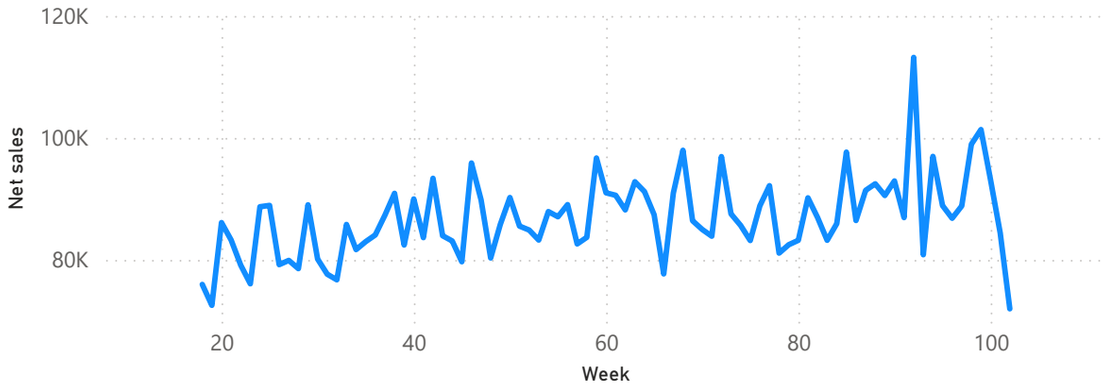
dunnhumby The Complete Journey: 2,500 households, 102 weeks. Weeks 1-17 (households still joining the panel) are filtered out.

Weeks

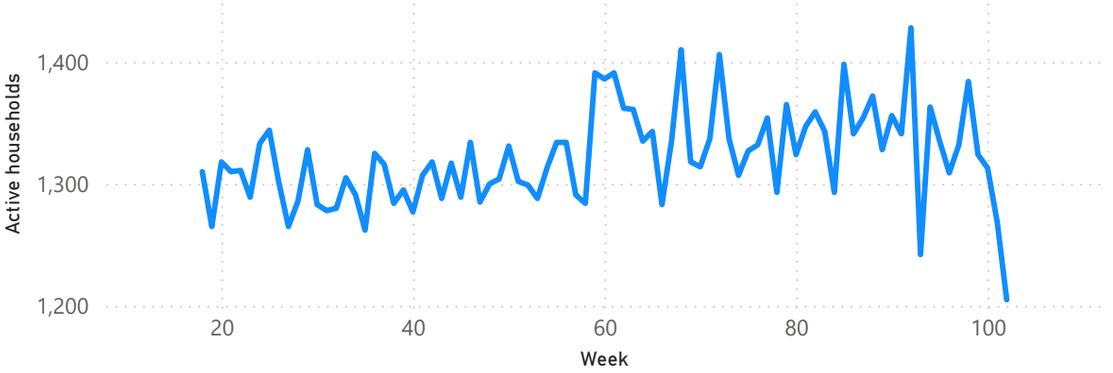
18 102

7.38M	1,318	29.33	89.8%
Net sales	Active households (median per week)	Basket value	4-week retention

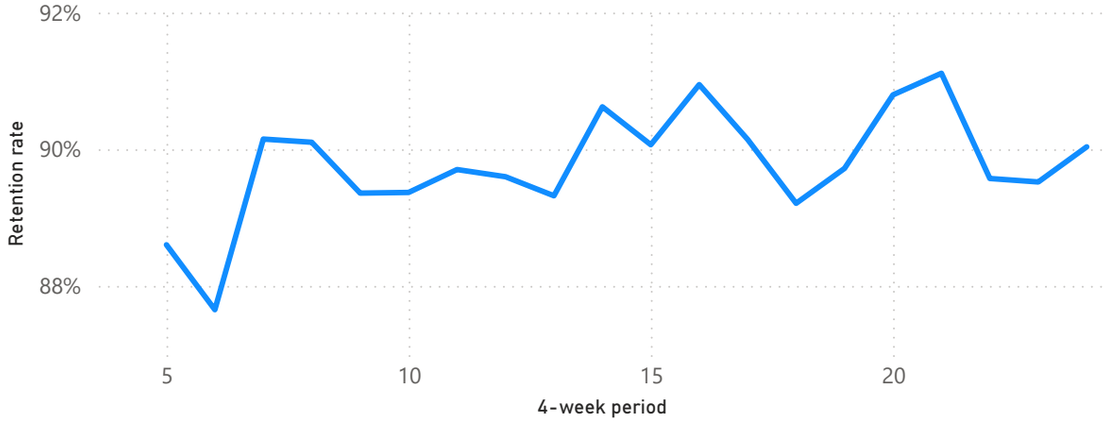
Net sales per week



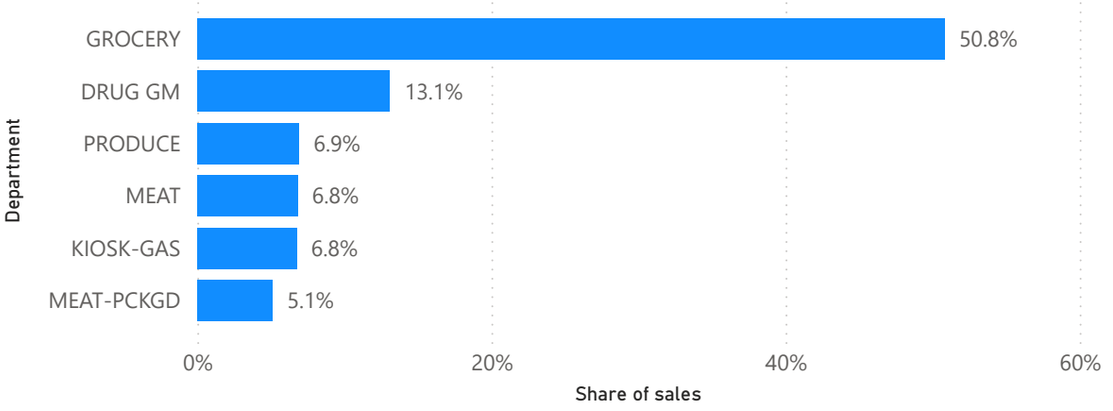
Active households per week



Retention: active again in the next 4-week period



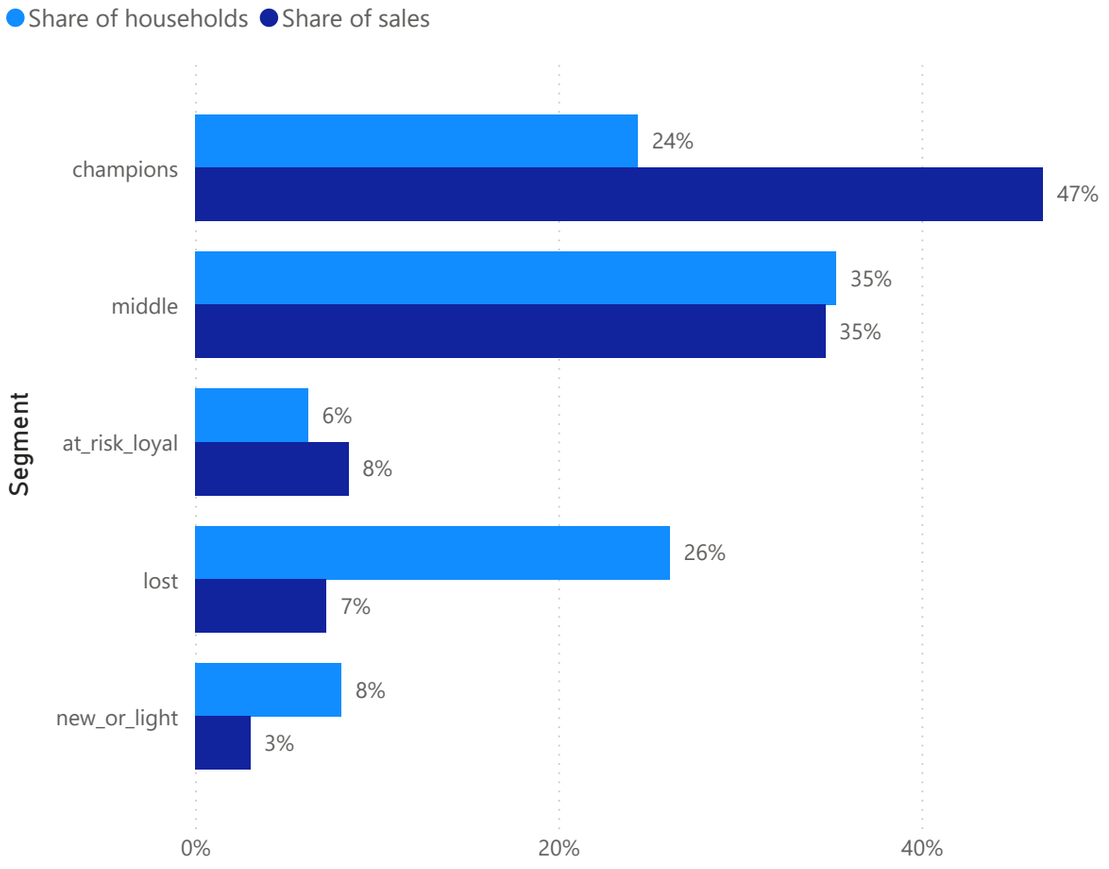
Top 6 departments by share of sales (all weeks)



Customer segments (RFM)

Champions are 24% of households but bring 47% of sales. Lost households are 26% of households and only 7% of sales.

Share of households vs share of sales



Segment table (weeks are medians)

Segment	Households	Sales / household	Active weeks	Weeks since last buy
champions	609	6,173	77	0
at_risk_loyal	155	4,380	67	2
middle	882	3,168	51	1
new_or_light	201	1,209	28	0
lost	653	891	22	5
Total	**2,500**	**3,223**		

How to read it

RFM = recency, frequency and monetary value over the whole 102 weeks (sql/02_rfm.sql). The segments describe customers; the churn model does not use them, because they contain the future.

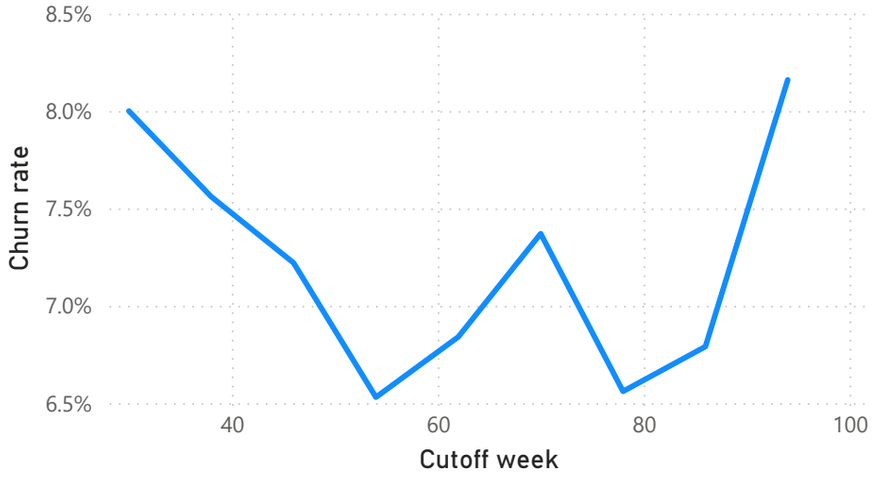
Losing one champion costs far more than losing one light shopper: the coupon list ranks households by expected sales lost, not by churn risk alone.

Churn & coupon list

Churn = no purchase in the next 8 weeks, among households that bought in the last 12 weeks. Coupon list = the 10% of at-risk households with the highest expected sales lost in week 102.

8.2%	+1.4	231	8,826
Churn rate (week 94)	Change vs week 86 (percentage points)	Households on the coupon list	Expected sales lost on the list

8-week churn rate per snapshot



How to use the list

Expected sales lost = churn probability x sales in the last 12 weeks (src/predict.py).
Whether a coupon really keeps a household is unknown: test it first. Give the coupon to a random half of the list and compare churn after 8 weeks (A/B test).

Coupon list (highest expected sales lost first)

Household	Churn risk	Sales (12 wk)	Expected loss	Weeks silent	Active weeks (12 wk)
149	18.9%	753	142.6	8	4
1016	25.3%	508	128.6	8	2
2042	22.7%	467	106.0	4	2
1629	41.5%	228	94.8	7	1
910	30.7%	285	87.5	10	1
700	7.8%	1,092	84.9	1	7
2251	24.0%	340	81.7	7	2
1702	20.1%	402	80.8	8	3
1140	34.7%	215	74.7	7	2
2136	9.2%	805	74.3	1	6
1215	23.7%	307	72.8	2	2
2420	14.8%	485	71.8	6	5
474	7.0%	978	68.8	0	7
931	23.9%	284	68.0	6	2
1507	8.9%	690	61.5	1	7
2408	8.9%	674	59.9	2	5
1657	20.0%	297	59.3	4	4
2374	11.4%	520	59.2	4	5
1205	16.9%	349	59.0	4	7
1496	38.3%	153	58.7	7	1
237	42.9%	136	58.2	8	1